

MCGRATH, AK, USA

WMO: 702310

Lat: 62.957N

Lon: 155.610W

Elev: 333

StdP: 14.52

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26510

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.4	-39.9	-35.8	0.7	-28.9	-33.3	0.8	-27.2	18.6	17.4	16.4	11.9	0.5	280	0.688

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	78.8	60.7	75.1	59.0	71.8	57.7	62.8	74.2	61.0	71.4	59.4	68.6	6.0	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.8	74.9	65.0	56.8	69.7	63.4	55.2	65.6	62.0	28.5	74.2	27.1	71.4	26.1	68.8	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	13.8	11.8	-47.9	83.7	5.8	3.4	-52.1	86.1	-55.4	88.1	-58.7	90.0	-62.9	92.5		
(5)			-38.4	66.6	6.8	2.6	-43.3	68.5	-47.3	70.0	-51.1	71.5	-56.1	73.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	-6.0	5.0	11.5	31.8	47.9	58.0	60.1	55.2	45.8	28.2	7.5	-2.8		
	DBStd	26.69	20.44	16.55	13.60	10.34	7.57	5.17	4.92	5.46	6.59	10.93	15.11	17.27		
	HDD50	8625	1735	1259	1195	547	126	4	1	15	152	677	1276	1638		
	HDD65	13301	2200	1679	1660	997	531	218	165	306	575	1141	1726	2103		
	CDD50	821	0	0	0	0	60	244	313	176	27	1	0	0		
	CDD65	24	0	0	0	0	0	8	13	3	0	0	0	0		
	CDH74	417	0	0	0	0	25	156	189	47	0	0	0	0		
	CDH80	58	0	0	0	0	1	27	25	5	0	0	0	0		
	WSAvg	4.0	2.4	3.2	4.5	5.3	5.6	4.8	4.5	4.3	4.3	4.0	3.0	2.2		
	PrecAvg	16.9	0.9	0.7	0.7	0.8	0.9	1.6	2.3	2.6	2.3	1.5	1.4	1.3		
	PrecMax	25.7	3.7	3.7	2.8	2.8	3.1	3.5	5.4	5.4	6.2	4.6	4.3	5.0		
(15) Precipitation	PrecMin	7.0	0.1	0.0	0.0	0.0	0.1	0.2	0.4	0.2	0.2	0.2	0.0	0.1		
	PrecStd	4.1	0.8	0.7	0.7	0.7	0.6	0.8	1.2	1.0	1.6	0.9	1.0	1.1		
	DB	39.9	41.0	45.5	60.0	77.5	83.4	83.1	79.7	68.5	54.7	41.4	38.5			
	MCWB	34.6	35.4	36.7	44.9	55.2	60.6	63.0	63.7	54.4	47.0	37.1	33.9			
	2% DB	34.2	36.7	40.1	55.2	71.7	78.3	79.1	73.3	63.7	48.6	36.2	33.2			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	32.0	33.7	34.1	42.6	53.3	59.5	61.5	60.3	52.2	43.4	34.2	31.2			
	5% DB	30.1	32.7	36.4	51.3	67.1	74.5	75.4	69.0	60.3	45.1	32.8	27.3			
	MCWB	28.5	30.3	32.0	40.4	50.9	57.6	60.4	58.2	50.8	41.2	31.5	25.7			
	10% DB	22.0	27.8	32.5	47.6	62.7	71.0	71.8	65.4	56.9	41.9	28.2	21.4			
	MCWB	21.1	25.7	28.8	38.6	48.8	56.2	59.3	56.2	49.8	38.8	26.9	20.5			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	35.7	37.3	37.9	46.3	56.8	64.7	65.5	65.7	55.9	48.0	37.9	34.7			
	MCDB	38.7	39.7	42.7	57.1	73.3	78.2	78.7	76.1	63.8	53.7	40.0	37.2			
	2% WB	32.8	33.9	35.0	43.4	54.5	61.2	63.3	62.0	53.9	44.2	34.2	31.5			
	MCDB	34.7	36.7	39.4	53.3	68.9	74.5	75.2	70.8	60.8	47.7	36.2	34.0			
	5% WB	29.4	30.5	32.4	41.2	52.1	59.2	61.6	59.7	52.1	41.6	31.6	26.2			
(31) Mean Daily Temperature Range	MCDB	31.4	33.1	36.3	50.3	64.1	71.5	72.5	66.7	58.2	44.9	32.8	28.2			
	10% WB	23.3	26.0	28.8	39.1	50.1	57.4	60.1	57.7	50.6	38.9	27.2	21.1			
	MCDB	24.6	28.1	32.5	47.0	61.0	68.3	69.7	63.4	56.0	41.6	28.4	22.1			
	5% DB	16.1	19.7	24.7	21.0	20.6	19.7	17.0	15.9	15.9	12.3	13.6	15.1			
	MCDBR	17.2	17.7	22.9	25.4	28.7	27.3	25.9	23.3	23.3	14.7	11.1	15.8			
(36) Clear-Sky Solar Irradiance	MCWBR	14.8	14.9	17.6	14.6	12.8	11.7	11.4	11.9	13.2	10.0	9.9	14.0			
	5% WB	16.8	17.5	19.9	22.9	25.6	23.7	22.0	19.5	19.0	13.1	11.0	15.9			
	MCDBR	14.4	14.9	15.5	13.7	12.0	10.9	10.6	10.5	11.4	9.2	9.9	14.1			
	taub	0.200	0.260	0.269	0.319	0.357	0.373	0.372	0.372	0.312	0.284	0.213	0.188			
	taud	2.028	2.217	2.272	2.229	2.378	2.355	2.405	2.415	2.563	2.445	2.027	1.914			
(42) All-Sky Solar Radiation	Ebn at Noon	146	221	268	273	272	269	266	255	255	214	131	74			
	Edh at Noon	10	19	26	35	33	35	32	29	21	16	10	5			
	RadAvg	80	309	884	1462	1641	1684	1442	1100	726	312	106	34			
(45)	RadStd	10	30	82	110	161	115	127	111	101	38	17	3			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			+1.80	N/A	+1.53	N/A	N/A	N/A	-538	-644	N/A
(47) Regional (5 neighbors)			+1.80	N/A	+1.53	N/A	N/A	N/A	-538	-644	N/A

Nomenclature: See separate page